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Разработка эмоционально-аннотированного видеодатасета дипфейков и исследование влияния динамики лицевых эмоций на устойчивость моделей обнаружения дипфейков

Topic

Development of an Emotion-Annotated Deepfake Video Dataset and Study of the Impact of Facial Emotional Dynamics on Deepfake Detection Robustness

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Abstract

Deepfake detectors generalize poorly across datasets because generation methods evolve faster than evaluation benchmarks. This thesis investigates whether facial emotional dynamics contribute to that gap and whether emotion-derived descriptors can improve detection robustness.

An emotion-annotated subset of Celeb-DF++ was constructed (1,000 videos across three forgery families and 22 generation methods, identity-disjoint splits) and processed through a pipeline of frame extraction, face detection, EMONET-FACE Large emotion annotation, video-level descriptor aggregation, and gated fusion training. Three pretrained detectors of different architectural complexity were evaluated under GroupKFold cross-validation grouped by subject identity: a HuggingFace image classifier, XceptionNet, and UCF (Uncovering Common Features).

Standalone detector AUC ranged from 0.540 to 0.730 on the held-out test partition. Detector fake scores correlated positively with maximum arousal ($\rho = +0.203$, $p = 8.6 \times 10^{-9}$): higher peak arousal makes detection easier, not harder, the opposite of the original hypothesis. A controlled decomposition revealed that quality features alone reach AUC near 0.90 across all three architectures, dominated by InsightFace detection confidence. After controlling for quality, emotional features retained significant independent signal on two of three architectures.

A three-modality gated fusion architecture combining quality, static emotional, and temporal emotional descriptors through a learnable per-video gating mechanism achieved test AUC 0.955 (95% CI 0.924–0.979) versus 0.730 for

UCF alone (DeLong $p = 1.8 \times 10^{-7}$), improving over UCF on 20 of 22 generators with the largest gain on the diffusion-based FLOAT generator (+0.414). The gating mechanism reveals method-specific strategies: quality dominates for face-reenactment and talking-face, static emotion for face-swap.

The main methodological implication is that quality-feature baselines should be standard in emotion-based deepfake detection research. The thesis contributes the emotion-annotated subset, a reproducible multi-architecture evaluation pipeline, an interpretable three-modality gated fusion architecture, and formal statistical testing throughout.

Keywords: deepfake detection, facial emotion recognition, valence-arousal, three-modality gated fusion, Celeb-DF++, EMONET-FACE Large, quality artifacts.

Chapter 1

Introduction

1.1 Background and Context

Deepfakes a shorthand for media generated or altered through deep neural networks first appeared on public platforms in 2017. The term itself combines “deep learning” and “fake.” Since their emergence, deepfakes have become substantially harder to detect and easier to produce [1, 2]. The dominant generation architectures are Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and, more recently, Diffusion models [2, 3].

Deepfakes serve a range of legitimate purposes, including visual effects in film and television, automated language dubbing, accessibility tools, and educational media that reconstructs historical figures or events. The same technology, however, has been used as a weapon for financial fraud, fabrication of evidence in legal and journalistic contexts, production of non-consensual explicit content, and political manipulation [2, 4, 5]. The proliferation of mobile applications and open-source frameworks has further reduced the technical barrier to creation, allowing non-experts to produce visually convincing face swaps. This widespread accessibility complicates detection efforts for both human reviewers and automated systems [4, 6].

Beyond the question of detection itself lies a more fundamental challenge:

generalization. Detectors that achieve AUC above 0.99 on benchmark datasets such as FaceForensics++ often experience substantial performance drops, on the order of 20 to 60 percentage points, when evaluated on out-of-distribution data [7, 8]. This gap between in-domain and real-world performance motivates the central question of this thesis: which features remain robust when the distribution shifts?

1.2 Problem Statement

Deepfake detection research has a methodological blind spot: models are evaluated on aggregate metrics over entire test sets, with no stratification by the emotional state of depicted subjects. This prevents direct measurement of whether, or how much, facial emotional dynamics affect detection difficulty.

Several bodies of evidence make this gap worth investigating. Deepfake generation methods are known to struggle with micro-expressions and rapid emotional transitions [1, 9, 10]. Prior work on emotion-based detection including the FEAD-D system [9] and more recent multimodal frameworks [5, 11] shows that emotional inconsistencies carry detectable signal. Yet major benchmark datasets including Celeb-DF++ [12] provide no emotion-level annotation, so stratified analysis of detection difficulty by emotional state has not been possible.

This leaves three questions unanswered in the existing literature:

1. **How does emotional state modulate the performance of pretrained deepfake detectors?**
2. **Are emotional features genuinely complementary to existing detector outputs, or do they overlap with technical quality artifacts that already drive detection?**
3. **Do emotion-based contributions generalize across detector architectures?**

This thesis addresses all three through stratified evaluation, quality-versus-emotion ablation, and multi-architecture comparison.

1.3 Research Goal

The goal of this work is to investigate the impact of facial emotional dynamics on the robustness of deepfake detection models, to develop an emotion-annotated deepfake video dataset, and to design an interpretable multimodal fusion architecture that quantifies which signals drive detection on which videos.

1.4 Research Objectives

The specific objectives are:

1. Analyze existing deepfake datasets and identify their limitations regarding emotional annotation.
2. Construct an emotion-annotated subset of the Celeb-DF++ benchmark with frame-level emotion predictions and aggregated video-level emotional descriptors, balanced across three forgery families and 22 generation methods.
3. Evaluate three pretrained deepfake detectors of different architectural complexity under emotion-stratified subgroup conditions, using identity-disjoint cross-validation.
4. Analyze detector errors and identify vulnerabilities with respect to specific emotional conditions and forgery method families.
5. Investigate whether adding emotional descriptors improves detection robustness, with controlled ablation distinguishing emotional signal from technical quality artifacts.

6. Design and evaluate an interpretable three-modality gated fusion architecture that separates quality, static emotional, and temporal emotional signals through a per-video gating mechanism.

1.5 Research Questions and Hypotheses

The primary research questions guiding this investigation are:

RQ1: Does the type and intensity of facial emotion affect deepfake detection quality?

RQ2: Which emotional conditions are the most difficult for deepfake detectors?

RQ3: Can emotional descriptors improve deepfake detection robustness, and to what extent are emotional features complementary to technical quality artifacts?

To operationalize these questions, this thesis tests the following hypotheses:

H1: Detection quality varies systematically across emotional conditions.

H2: Detection errors correlate with facial emotional dynamics.

H3: Adding emotional descriptors to detector outputs improves robustness, and at least part of this improvement remains after controlling for technical quality features.

Each hypothesis is tested using formal statistical methods: H1 through subgroup AUC comparison with DeLong’s test and bootstrap confidence intervals; H2 through Spearman rank correlation between detection error and video-level emotional descriptors; H3 through DeLong’s test and permutation tests comparing fusion against baseline detector performance, with explicit ablation against a quality-feature baseline to control for technical confounders.

1.6 Contributions

This thesis makes four contributions to deepfake detection and emotion-aware media analysis.

1. **An emotion-annotated subset of the Celeb-DF++ benchmark** (1,000 videos: 800 main + 200 pilot) with frame-level annotations from EMONET-FACE Large [13], providing continuous scores across 40 emotion categories per frame plus derived valence and arousal via Russell-circumplex mapping [14]. The subset is balanced across three forgery families and 22 generation methods, with identity-disjoint splits.
2. **A reproducible multi-stage pipeline** covering frame extraction, face detection via InsightFace, emotion annotation, video-level descriptor aggregation, baseline detector inference, and fusion classifier training with identity-disjoint GroupKFold cross-validation. The pipeline is evaluated across three pretrained detectors of varying complexity: HuggingFace, Xception-Net [15*], and UCF [16].
3. **A controlled decomposition of emotional contribution from technical quality artifacts.** On Celeb-DF++, quality features alone achieve AUC near 0.90 across all three detector architectures, exceeding the standalone detectors and accounting for a substantial portion of any apparent emotional contribution. After controlling for quality, emotional features retain statistically significant residual signal on HuggingFace and UCF but not on XceptionNet, establishing that emotion provides complementary information rather than acting purely as a proxy for technical artifacts.
4. **A three-modality gated fusion architecture** that separates quality, static-emotional, and temporal-emotional signals into three branches with a per-video learnable gating mechanism. The model achieves test AUC 0.955

(95% CI 0.924–0.979) compared to 0.730 for the UCF baseline alone—a $\Delta = +0.227$ improvement (DeLong $p = 1.8 \times 10^{-7}$). On 22 deepfake generation methods, the model improves over UCF on 20 of 22 generators (mean $\Delta = +0.155$, with the largest gain +0.414 on the diffusion-based FLOAT generator). The gating weights reveal which modality dominates on which video, providing per-sample interpretability: quality dominates for FaceReenact and TalkingFace, static emotion dominates for FaceSwap.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews deepfake generation, detection, facial emotion recognition, and their interaction, structured around the central question of whether facial emotional dynamics measurably affect the robustness of pretrained detectors. Section 2.2 covers generation techniques relevant to the Celeb-DF++ benchmark. Section 2.3 reviews detection methods, with emphasis on pretrained architectures and unified benchmarking frameworks. Section 2.4 introduces facial emotion recognition as a temporal signal. Section 2.5 examines prior emotion-based detection work. Section 2.6 reviews benchmarks and the evidence that quality artifacts drive much of reported detector performance. Section 2.7 covers statistical evaluation methods. Section 2.8 states the specific gap this thesis addresses.

2.2 Deepfake Generation: Modern Landscape and Methods

Deepfake generation has changed rapidly since consumer-accessible face-swap tools appeared in 2017. The dominant generative families used in current

deepfake pipelines are Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Diffusion models [2, 3, 17, 18]. For the purposes of this thesis, what matters is not the algorithmic details of each family but the distinct types of artifacts each produces, since artifact diversity is what makes cross-method detection difficult.

2.2.1 GANs and the Face-Swap Paradigm

GAN-based methods were the dominant approach in early deepfake generation and remain widely used. A generator network learns to produce synthetic faces under an adversarial signal from a discriminator trained to distinguish real from generated content [2]. Early systems struggled with pose and lighting consistency. Subsequent improvements came from several directions: high-quality face synthesis architectures such as StyleGAN [19] and its successors, integration of 3D Morphable Models [20] to constrain pose and identity, and latent-space manipulation techniques for attribute control. Open-source frameworks, most notably DeepFaceLab [18], made high-quality face-swap generation accessible outside specialist research groups and remain the predominant tool in publicly available deepfake datasets.

2.2.2 VAE-Based Face Replacement

VAE-based methods perform face replacement through latent-space recombination: two autoencoders with shared encoders but distinct decoders are trained on different identities, so that face A can be reconstructed using the decoder trained on face B [2]. VAE-only approaches have largely been superseded by GAN- and diffusion-based methods, though their architectural pattern persists in many hybrid systems.

2.2.3 Diffusion-Based Generation

Diffusion models are now the leading approach in image synthesis and have been adapted for deepfake generation, particularly for talking-face and face-reenactment scenarios [2, 3, 17]. These models progressively corrupt training data with noise in a forward pass and learn to reverse this process during inference. Compared with earlier GAN-based systems, diffusion models tend to produce smoother, higher-fidelity outputs with fewer visible artifacts [3], which makes them harder to detect.

2.3 Deepfake Detection: Methods and Benchmarks

Detection methods fall broadly into three generations: artifact-based CNN classifiers, temporal models that exploit inter-frame inconsistencies, and architectures incorporating semantic or biological cues [4, 6, 21, 22].

2.3.1 CNN-Based Detectors

XceptionNet, developed for general image classification [15*], became the standard detection baseline after Rössler et al. [23] showed it transferred to FaceForensics++. Much of the subsequent work stayed within this CNN paradigm: Meso-4 and MesoInception-4 [21] targeted mesoscopic image properties, EfficientNet-based ensembles followed [1, 24], and later multi-attention designs [3] focused attention more explicitly on facial subregions. The underlying logic is consistent across these approaches: generation artifacts tend to cluster around the eyes and mouth [9], and models trained to catch them there tend to work.

2.3.2 Disentanglement-Based Detection

One active line of work augments CNN classifiers with auxiliary objectives that separate forgery-relevant features from identity- or content-related features. The UCF (Uncovering Common Features) detector [16] adds a feature-disentanglement objective to an XceptionNet backbone, isolating forgery-specific signals from content-irrelevant variation. UCF ranked among the top-performing image-level detectors in the DeepfakeBench evaluation [16], particularly in within-domain settings, which motivated its inclusion in the present work alongside vanilla XceptionNet.

2.3.3 Temporal and Spatio-Temporal Detection

Frame-level CNNs discard temporal information that can reveal generation artifacts. Recurrent architectures such as LSTM and Bidirectional LSTM models address this by capturing inter-frame inconsistencies [3]. More recent spatio-temporal architectures jointly encode spatial and temporal features, with several studies reporting improved cross-method generalization [3, 25]. Related work explores complementary cues — frequency-domain features and self-supervised pretraining among them — to identify signals that transfer across generation pipelines [3].

2.3.4 Benchmark Frameworks

Reported AUC values in deepfake detection often vary substantially due to differences in preprocessing, splits, frame sampling, and post-processing. DeepfakeBench [16] addressed this by standardizing training and evaluation across 34 detection methods. The present work uses XceptionNet and UCF from DeepfakeBench, leveraging their publicly released pretrained checkpoints as baselines.

2.3.5 Cross-Dataset Generalization

Deepfake detectors generalize poorly across datasets, a finding consistent across multiple recent reviews [3, 4, 6, 22]. Reported performance drops vary with the distance between training and evaluation distributions. In the worst cases, detectors achieving high AUC on FaceForensics++ lose substantial discriminative power on out-of-distribution data [7]. Even on closely related benchmarks, the drop is non-negligible: the Celeb-DF++ benchmark [12] reports a measurable mean frame-AUC reduction when detectors trained on FaceForensics++ are evaluated on it. Akhtar et al. [4] attribute this fragility to overfitting on generator-specific artifacts rather than learning transferable manipulation cues.

2.4 Facial Emotion Recognition in Video

In video, facial emotion is a temporal signal: it unfolds across onset, peak, and offset phases, with asynchronous and non-linear movements of facial features [1]. Computational approaches need to treat it as such rather than as a static per-frame classification.

2.4.1 Categorical and Dimensional Models

FER systems use two complementary representations. **Categorical models** assign each face to one of a fixed set of discrete emotion classes, most commonly the six basic emotions identified by Ekman [26]: anger, disgust, fear, happiness, sadness, and surprise. Modern FER datasets typically extend this set with neutral and sometimes contempt. **Dimensional models** represent emotion as a continuous point in affective space, most commonly along valence-arousal axes, where valence captures positive-negative polarity and arousal captures activation intensity [14]. Dimensional representations capture intensity gradations and emotional change over time in ways that categorical labels cannot.

2.4.2 Pretrained Emotion Models Used in This Work

EMONET-FACE Large [13] is the primary annotation model used in this work. Trained on the EMONET-FACE benchmark suite, it produces 40 continuous emotion category scores per frame. Valence and arousal are derived from the 40-category outputs via a Russell-circumplex mapping [14]. The model provides fine-grained annotation with explicit demographic balancing across gender, age, and 14 ethnic groups, addressing the Western-centric bias common in earlier FER datasets.

2.5 Emotion-Based Deepfake Detection

The basis for using emotional inconsistency as a detection signal is that generative models struggle to reproduce realistic emotional dynamics. They tend to smooth or average emotional transitions, producing facial motion that is more linear and synchronous than natural expression [1, 9].

2.5.1 Foundational Work on Emotion-Based Deepfake Detection

Several early studies showed that emotional cues can expose deepfake manipulation. Mittal et al. [27] compared perceived emotion in facial expressions against perceived emotion in speech and found that deepfakes often show mismatches between the two. López-Gil et al. [10] tested how well those expressions transfer from originals to face-swap deepfakes on Celeb-DF; deepfakes rarely preserved them accurately, especially for complex emotions like fear and anger. Hosler et al. [28] took a semantic angle and found detectable inconsistencies in emotional content over time. Gravina et al. [9] then built FEAD-D, putting facial expression analysis directly into the detection pipeline. The common thread: emotion-derived signals carry information that pixel-level features miss.

2.5.2 Architectural Approaches

FEAD-D [9] demonstrated that emotion-based detection can be competitive with appearance-based methods at lower computational cost. Its three-module pipeline face detection, textural and emotional feature extraction, and Bidirectional LSTM temporal analysis reached 84.29% accuracy on the DFDC test set with substantially fewer parameters than competing approaches. Anand et al. [29] explored a finer-grained alternative, using facial Action Units to guide detection of localized manipulations such as subtle changes in eyebrows, eye shape, or mouth movements that global features tend to miss.

2.5.3 Recent Developments

Two recent directions illustrate where the field is moving. Lee and John [5] took a human-perception perspective, showing that the emotional expressiveness lost during face-swap deepfake generation correlates with reduced credibility for viewers linking algorithmic detectability to perceptual judgment. Zhang et al. [11] took a model-centric route with their Emoencoder framework, which uses emotion-aware contrastive learning to exploit cross-modal emotional inconsistencies and target generalization to unseen identities in few-shot settings.

2.5.4 Limitations of Existing Emotion-Based Work

Three methodological limitations recur in this literature. First, prior studies have not disentangled emotional features from co-varying technical signals such as generation-pipeline quality artifacts; without controlled ablation against quality baselines, an apparent emotion contribution may reflect overlap with technical confounders. Second, evaluations are typically conducted on a single detector architecture, so it is unclear whether emotion-based improvements generalize across architectures or depend on specific feature representations. Third, statistical eval-

uation often relies on point estimates without formal hypothesis testing or out-of-sample validation. This thesis addresses all three.

2.6 Datasets and the Quality Artifact Concern

Benchmark dataset choice shapes what detectors learn and what conclusions can be drawn from their performance [4, 21, 24].

2.6.1 Major Benchmark Datasets

FaceForensics++ [23] (1,000 real videos, 4,000 manipulated with four classic methods) remains the most widely cited benchmark despite its limited method coverage. **Celeb-DF** [30] introduced higher-quality encoder-decoder forgeries but skews 70/30 toward male subjects. **Celeb-DF++** [12], the primary benchmark used in this thesis, expands Celeb-DF to 53,196 fake clips produced by 22 generation methods across face-swap, face-reenactment, and talking-face scenarios, and defines three evaluation protocols (GF-eval, GFQ-eval, GFD-eval) for cross-method, cross-compression, and cross-dataset transfer. Other widely used benchmarks include **DFDC** [9], which provides over 100,000 videos with strong demographic diversity, and **AV-Deepfake1M** [24], which contributes content-driven audio-visual manipulations. None of these benchmarks provides systematic emotion-level annotation, motivating the annotated subset constructed in this thesis.

2.6.2 Demographic Bias

Nadimpalli and Rattani [21] demonstrated that demographic imbalance in FaceForensics++ leads to unequal detector performance across gender groups. Celeb-DF++ inherits the demographic skew of its real-video pool (approximately 88% Caucasian, 7% African-American, and 5% Asian [30]), constraining the populations to which conclusions from any Celeb-DF++-based study can be general-

ized. FER models trained on Western-centric data compound this issue: annotation biases propagate to downstream emotion-based analyses [31, 32*].

2.6.3 The Quality Artifact Concern

Deepfake generation pipelines produce characteristic patterns in face detection confidence, temporal stability, sharpness, and compression behavior. These technical artifacts differ systematically between real and fake videos and can drive detector decisions independently of any semantic content. Recent work has flagged this as a source of spurious generalization: Song et al. [8] argued that detectors trained on mixtures of low- and high-quality forgeries exploit artifact shortcuts; Ma et al. [33] distinguished generator-specific artifacts from generalizable manipulation cues; and Liu et al. [3] noted that binary classifiers can learn dataset-specific identity traits rather than manipulation cues. For emotion-based detection in particular, any apparent contribution from emotional features must be evaluated against a quality-feature baseline to distinguish genuine semantic signal from incidental overlap with technical artifacts. The present work addresses this through controlled quality-versus-emotion ablation.

2.7 Statistical Methods for Detector Evaluation

Point estimates of AUC, accuracy, or F1 are insufficient to distinguish real performance differences from sampling noise on small test sets common in deepfake research. This thesis combines bootstrap confidence intervals [34], DeLong’s test for paired AUC comparison [35], Spearman rank correlation [36] for monotonic associations, and permutation tests for distribution-free model comparison. Identity-disjoint cross-validation via GroupKFold, grouped on subject identity, prevents the inflation that results from a detector memorizing person-specific appearance.

2.8 Research Gap

The literature shows that emotional dynamics differ measurably between real and synthesized facial videos and can in principle be used for detection [1, 5, 9, 10, 11]. Three concrete gaps remain. No prior work has systematically evaluated how emotional state modulates detector performance across stratified emotional conditions and multiple architectures. The relative contribution of emotional descriptors versus technical quality artifacts has not been decomposed, despite mounting evidence that quality artifacts dominate detector decisions on benchmark datasets [7, 8, 33]. And prior emotion-based studies have relied on point estimates without formal hypothesis testing, identity-disjoint validation, or out-of-sample checks. This thesis addresses all three through the methodology in Chapter 3, the results in Chapter 4, and the discussion in Chapter 5.

Chapter 3

Design and Methodology

3.1 Dataset Curation

The primary benchmark selected for this study was Celeb-DF++ [12], chosen because the objectives of this work require substantial diversity in forgery generation methods together with video quality high enough for reliable facial-emotion analysis. Unlike earlier benchmarks such as Celeb-DF [30], which primarily focuses on a single face-swap pipeline, Celeb-DF++ contains more than 53,000 fake videos generated by 22 modern manipulation methods spanning face-swap, face-reenactment, and talking-face scenarios, along with dedicated evaluation protocols for cross-method, cross-compression, and cross-dataset robustness—making it particularly suitable for studying detector generalization under heterogeneous generation conditions. FaceForensics++ [23] was also considered during benchmark selection due to its widespread adoption in deepfake detection research, but its limited manipulation diversity and focus on earlier-generation forgery methods made it less aligned with the robustness-oriented objectives of this study. Several known limitations of Celeb-DF++ were considered during result interpretation, including demographic imbalance in the real-video pool, limited compression diversity, and the exclusion of forgery categories such as full-face synthesis and facial-attribute editing.

3.1.1 Curated Subset Construction

A curated subset was constructed instead of using the full benchmark in order to support detailed emotion-stratified analysis under realistic computational constraints. Videos were included only if a primary face was visible and trackable for most frames, the aligned face crop exceeded 256×256 pixels, the clip duration was at least three seconds, severe compression artefacts or unstable tracking were absent, and fewer than 20% of sampled frames lacked a detectable face.

The final dataset consisted of two subsets: a pilot subset of 200 videos for pipeline validation and final out-of-sample generalization checks, and a main subset of 800 videos for all reported training and evaluation experiments. The combined dataset therefore contains 1,000 videos balanced across real and fake classes and approximately balanced across forgery families and generation methods. The main subset was divided identity-disjointly into an 80% trainval partition (635 videos) and a 20% test holdout partition (155 videos).

Five-fold cross-validation on the trainval partition was performed using GroupKFold grouped by subject identity to prevent identity leakage between training and validation folds. Figure 3.1 shows the resulting class and forgery-method distribution across the combined 800-video main subset and 200-video pilot subset.

3.2 Face Detection and Alignment

Frames were extracted from each video at 5 frames per second (fps), balancing temporal coverage with storage and compute requirements. Face detection and alignment were performed using InsightFace RetinaFace with the `buffalo_l` model bundle. Detection used the `det_10g.onnx` model at input resolution 640×640 with CUDA-accelerated ONNX inference. Five-point facial landmarks (eyes, nose, mouth corners) were extracted from the detection output and

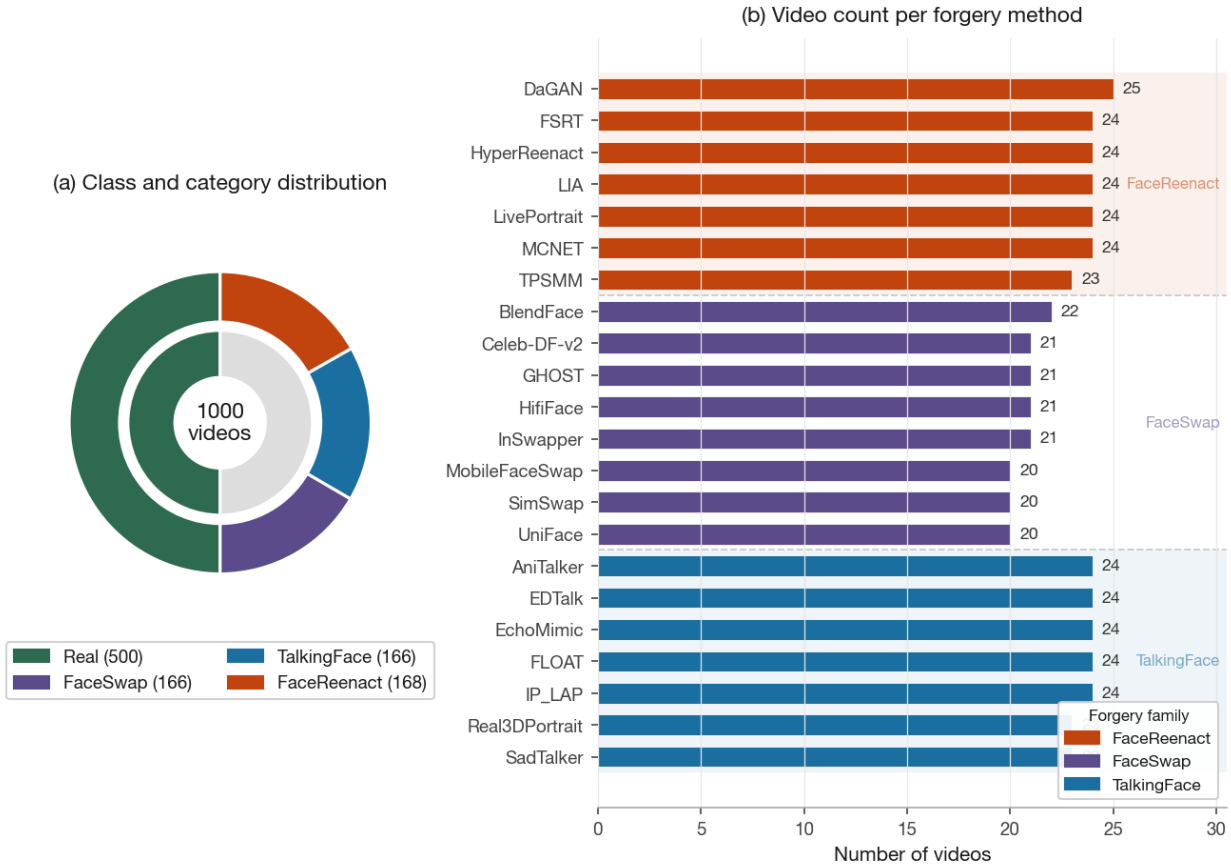


Figure 3.1: Class and method distribution of the curated subset.

used to align each detected face crop to a canonical frame. RetinaFace was selected because it provides robust performance on partially occluded and low-resolution faces while maintaining high-quality landmark precision for downstream alignment. For each frame, only the largest detected face was retained and aligned using the estimated landmarks. Frames without a detectable face were discarded. The resulting aligned crops formed the input to the emotion annotation stage.

3.3 Frame-Level Emotion Annotation

Frame-level emotion predictions were extracted using EMONET-FACE Large [13], which combines a SigLIP-2 vision encoder [37*] with 40 regression heads, one per fine-grained emotion category. The model was selected because it produces continuous emotion scores rather than discrete labels, supports dimensional affect analysis through valence and arousal derivation, and was trained on



Figure 3.2: Face detection and alignment pipeline.

data with substantially improved demographic balance relative to earlier facial-expression-recognition datasets.

For each aligned face crop, the model outputs 40 continuous scores. Valence and arousal were derived from these scores using a fixed Russell-circumplex mapping [14]:

$$\text{valence}_{\text{frame}} = \frac{\sum_{e=1}^{40} \max(s_e, 0) \cdot v_e}{\sum_{e=1}^{40} \max(s_e, 0)} \quad (3.1)$$

$$\text{arousal}_{\text{frame}} = \frac{\sum_{e=1}^{40} \max(s_e, 0) \cdot a_e}{\sum_{e=1}^{40} \max(s_e, 0)} \quad (3.2)$$

where s_e denotes the predicted score for emotion category e and (v_e, a_e) are the predefined Russell-circumplex coordinates.

3.4 Video-Level Descriptor Aggregation

Frame-level predictions were aggregated into fixed-length video-level descriptors to support classifier training and subgroup analysis. Three descriptor groups were constructed.

3.4.1 Static Emotional Descriptors

Static descriptors summarize average emotional content across all frames in a video. These include: per-category emotion means (computed from the 40 emotion dimensions); dominant emotion one-hot encoding (indicating the emotion category with highest mean score); mean valence and mean arousal derived from the Russell-circumplex projection; and neutral-expression ratio (proportion of frames with neutral as the dominant emotion).

3.4.2 Temporal Emotional Descriptors

Temporal descriptors capture emotional dynamics across video frames. These include: per-category standard deviation and frame-to-frame temporal gradient (measuring frame-level variability and trends); arousal variation (standard deviation of frame-level arousal); transition rate between dominant emotions (frequency of switches to a new dominant emotion); and emotion entropy (Shannon entropy of the dominant emotion distribution, measuring emotional diversity over time).

3.4.3 Technical Quality Descriptors

Technical descriptors were additionally computed to separate emotional effects from dataset-specific artefacts and capture preprocessing quality. These include: face-detection confidence statistics (mean, min, and max detection confidence across frames); frame stability (computed as mean interframe optical-flow magnitude); sharpness statistics (Laplacian-based blur estimate); frame count (total frames per video); and aggregate detector-score statistics from the deepfake detector baseline (described in Section 3.5). The separation into quality, static-emotional, and temporal-emotional groups directly supports the multimodal fusion design introduced later in this chapter.

3.5 Baseline Deepfake Detection

Three pretrained deepfake detectors were evaluated. The primary baseline was UCF (Uncovering Common Features) [16], a feature-disentanglement detector implemented within the DeepfakeBench framework. The second was XceptionNet [15*], the standard CNN architecture commonly used in deepfake detection research. The third was a publicly available HuggingFace detector [38], included as a lightweight off-the-shelf reference.

Each detector produced frame-level probabilities averaged into video-level authenticity scores. Evaluation was performed both globally and across emotion-defined subgroups, with area under the ROC curve (AUC) as the primary metric.

3.6 Fusion Architecture

3.6.1 Initial Fusion Baselines

Initial experiments combined detector outputs with aggregated emotional descriptors using logistic regression, linear SVM, and XGBoost classifiers [39]. These models established whether emotion-derived features contain signal beyond detector outputs alone.

3.6.2 Quality-versus-Emotion Decomposition

To separate emotional contribution from technical artefacts, quality and emotion features were evaluated independently and jointly under five-fold GroupK-Fold cross-validation. The decomposition addressed two questions: how much of the performance improvement is explained by quality artefacts alone, and whether emotional descriptors retain independent predictive signal after controlling for quality. Statistical significance was evaluated using DeLong’s test and permutation tests.

3.6.3 Three-Modality Gated Fusion Architecture

A custom three-modality gated fusion network was designed to model interactions between technical quality features, static emotional descriptors, and temporal emotional descriptors. As illustrated in Figure 3.3, each modality is processed by an independent multilayer perceptron (MLP) that produces a latent embedding and a modality-specific logit. A learnable gating head, also an MLP, computes

per-video softmax weights over the three branches.

The final prediction is defined as:

$$z_{\text{final}} = g_q z_q + g_s z_s + g_t z_t, \quad \hat{y} = \sigma(z_{\text{final}}) \quad (3.3)$$

where g_q , g_s , and g_t are the learned gating weights for the quality, static-emotional, and temporal-emotional modalities respectively.

3.6.4 Ablation Design

To evaluate each modality’s contribution, three ablation variants were trained: one without quality features, one without static emotional features, and one without temporal emotional features. Each was retrained independently under identical cross-validation conditions.

3.7 Evaluation Protocol

All experiments used identity-disjoint evaluation. Model development and cross-validation were performed exclusively on the main subset trainval partition, while the reserved test holdout was used only for final evaluation after model selection. The pilot subset was never used for training or hyperparameter tuning; instead, it served as an unseen out-of-sample generalization check. All analytical visualizations describing dataset statistics, emotion distributions, or modality behaviour were computed on the combined pilot and main datasets to avoid reporting distributions from a single partition only. The primary evaluation metric was AUC, supplemented by accuracy, precision, recall, F1-score, and equal error rate (EER). Statistical significance was assessed using bootstrap confidence intervals, DeLong’s test for AUC comparison, and permutation tests.

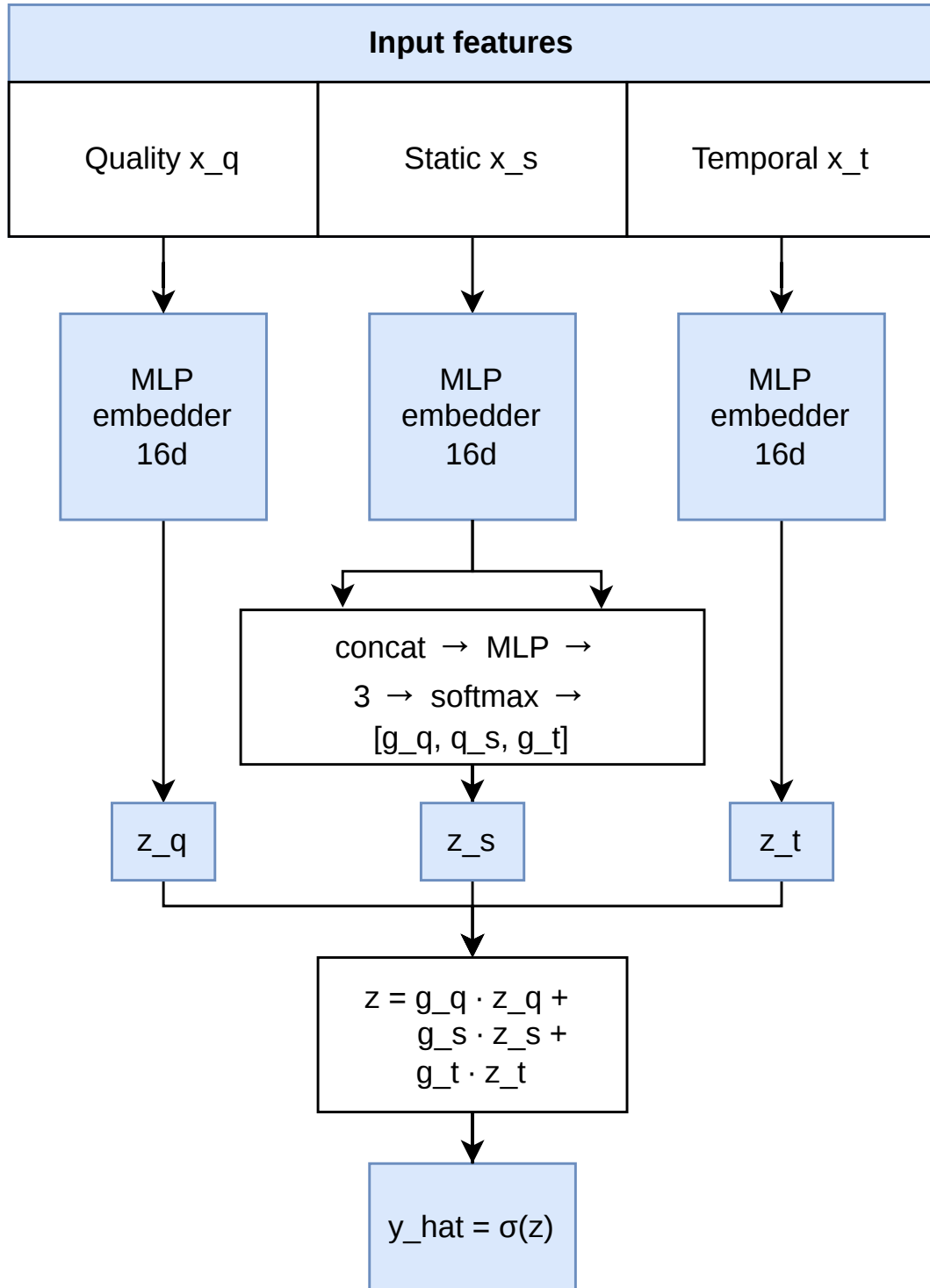


Figure 3.3: The three-modality gated fusion architecture.

Chapter 4

Implementation and Results

4.1 Implementation Environment

All experiments were executed on a remote GPU server equipped with an NVIDIA A100 GPU (80 GB VRAM) and 200 GB of SSD storage. Pipeline stages requiring deep-learning inference (face detection, emotion annotation, baseline detector inference) were executed on GPU; downstream stages (descriptor aggregation, fusion training, statistical testing) were executed on CPU. The software stack was based on Python 3.10, PyTorch, ONNX Runtime, scikit-learn, XGBoost, SHAP, and UMAP. A fixed random seed of 42 was used throughout to ensure reproducibility. Pretrained model weights were obtained from public sources: XceptionNet and UCF from the DeepfakeBench v1.0.1 release, EMONET-FACE Large from the LAION HuggingFace repository, and InsightFace buffalo_1 via automatic download.

Fifteen experiments (Exp01–Exp15) were conducted on the curated CelebDF++ subset ($N = 790$; 635 trainval, 155 test holdout; 62 identities; 3 forgery families; 22 generators). All experiments used identity-disjoint GroupKFold cross-validation grouped on subject identity. Throughout this chapter, AUC denotes the area under the Receiver Operating Characteristic curve. Confidence intervals were computed using 2,000-iteration bootstrap resampling; p -values were computed us-

ing DeLong’s test for AUC comparisons and permutation tests (10,000 iterations) for fusion-versus-baseline contrasts.

4.2 Baseline Detector Evaluation

Three pretrained detectors were evaluated without fusion to establish reference performance: a lightweight HuggingFace image classifier [38], XceptionNet, and UCF (both obtained through DeepfakeBench). Per-frame softmax probabilities were aggregated to the video level by mean pooling. Results are reported in Table 4.1.

TABLE 4.1
Standalone detector AUC (no fusion features).

Detector	AUC (all, $n=790$)	AUC (test, $n=155$)
HuggingFace	0.540	0.603
XceptionNet	0.674	0.674
UCF	0.757	0.730

UCF therefore served as the strongest standalone baseline in the present study and as the primary reference for the fusion experiments in Section 4.7.

As a diagnostic step, a four-condition logistic regression ablation was performed on XceptionNet to identify which emotional feature subset contributes most to fusion performance. Combining XceptionNet with affect features (valence, arousal, dominant emotion) improved test AUC from 0.674 to 0.721. Combining with dynamics features alone (arousal variation, transition rate, emotion entropy) degraded performance to 0.558, suggesting that under linear fusion at this sample size the dynamics descriptors require interaction with affect to be informative. XGBoost fusion was also tested and underperformed logistic regression on the test holdout (best AUC 0.675), confirming the linear classifier as the more reliable choice for this dataset size.

4.3 Subgroup Analysis under Stratified Emotional Conditions

Subgroup analyses in this section were performed across all three detectors to verify that observed patterns are not architecture-specific. The analyses use the full dataset ($n = 790$).

4.3.1 Per-emotion-class AUC

Each detector was stratified by the dominant emotion class predicted by EMONET-FACE Large on the full dataset. Of the 40 fine-grained emotion categories, ten contained sufficient samples ($n \geq 10$ with both real and fake examples) for reliable per-class AUC estimation. Results are reported in Table 4.2.

TABLE 4.2
Per-emotion AUC across three detectors ($n=790$).

Emotion class	n	HF	XceptionNet	UCF
amusement	14	0.675	0.950	0.950
interest	17	0.569	0.917	0.875
embarrassment	17	0.619	0.762	0.929
thankfulness_gratitude	11	0.222	0.722	0.944
elation	136	0.493	0.702	0.742
astonishment_surprise	263	0.488	0.689	0.762
contentment	282	0.578	0.670	0.718
sexual_lust	18	0.416	0.636	0.870
anger	20	0.625	0.583	0.802
disappointment	10	0.619	0.381	0.762

All three detectors show substantial AUC variation across emotion classes, confirming that detection difficulty depends on emotional content. The pattern of difficulty differs across architectures: XceptionNet performs poorest on disappointment (AUC 0.381) and best on amusement (0.950), while UCF achieves consistently higher performance across all classes (range 0.718–0.950, with a single outlier at disappointment 0.762). HuggingFace shows the most uneven pattern, with thankfulness_gratitude as a substantial weakness (AUC 0.222) even when stronger

detectors handle this class well (UCF 0.944). This supports the weaker formulation of hypothesis H1 that emotional condition modulates detection but the direction and magnitude of the modulation are architecture-dependent.

4.3.2 Forgery family $\times$ emotion interaction

To examine interaction effects between forgery type and emotional content, AUC was computed for each combination of forgery family and dominant emotion class on each detector. The cross-tabulation reveals that detection difficulty is determined by the combination rather than by either factor alone, and that the interaction pattern differs substantially across detectors.

For the HuggingFace baseline, FaceSwap is the easiest family (AUC 0.448–0.667 across emotions), while TalkingFace contains the single hardest cell (AUC 0.188 on thankfulness_gratitude). For XceptionNet, FaceSwap is by contrast the hardest family (AUC 0.200–0.562), while TalkingFace is the easiest (AUC 0.688–1.000). For UCF, all three families produce high AUC (0.661–0.944), with no single family consistently harder than the others. Selected cells are reported in Table 4.3.

TABLE 4.3
Selected forgery $\times$ emotion AUC cells across three detectors ($n=790$).

Forgery family	Emotion	HF	XceptionNet	UCF
FaceReenact	contentment	0.595	0.842	0.741
FaceReenact	astonishment_surprise	0.475	0.764	0.781
FaceReenact	elation	0.491	0.719	0.661
FaceSwap	disappointment	0.667	0.200	0.867
FaceSwap	contentment	0.604	0.439	0.667
FaceSwap	sexual_lust	0.500	0.476	0.857
FaceSwap	elation	0.531	0.250	0.804
TalkingFace	embarrassment	0.619	1.000	0.905
TalkingFace	contentment	0.500	0.871	0.786
TalkingFace	thankfulness_gratitude	0.188	0.688	0.938
TalkingFace	elation	0.482	0.839	0.762

The architectural disagreement is most pronounced on FaceSwap: Xception-

Net generalises poorly to the face-swap methods present in Celeb-DF++ (several cells at or below random), while UCF’s disentanglement objective recovers strong performance on the same combinations. This confirms that conclusions drawn from any single detector architecture are not transferable across detectors and motivates the multi-architecture decomposition in Section 4.7.

4.4 Statistical Hypothesis Testing

The three hypotheses formulated in Chapter 3 were tested separately on each of the three detectors. This multi-architecture testing was performed to verify that conclusions about emotion-detector interaction are not specific to a single architecture.

4.4.1 H1: arousal-stratified AUC

The dataset was stratified into low-, medium-, and high-arousal terciles using quantile-based binning on mean arousal. AUC per tercile was computed for each detector. Results are reported in Table 4.4.

TABLE 4.4
H1: AUC by arousal tercile across three detectors ($n=790$).

Detector	Low arousal	Medium arousal	High arousal
HuggingFace	0.545	0.518	0.518
XceptionNet	0.685	0.629	0.748
UCF	0.788	0.708	0.785

None of the three detectors exhibits the monotonically decreasing pattern predicted by the original formulation of H1. HuggingFace shows no meaningful tercile variation. XceptionNet and UCF both show a U-shaped pattern, with medium arousal harder than both low and high. The strict monotonic version of H1 is therefore not supported on any detector. A weaker version that detection quality varies

across emotional conditions is supported by the per-emotion AUC spread shown in Section 4.3.

4.4.2 H2: emotion-detector correlations

Spearman rank correlations between video-level emotion descriptors and detector fake scores were computed for each detector on the full dataset ($n = 790$). Results are reported in Table 4.5.

TABLE 4.5

H2: Spearman ρ between emotion descriptors and detector fake score, across three detectors ($n=790$). Significance: *** $p < 10^{-3}$, ** $p < 10^{-2}$, * $p < 0.05$, n.s. otherwise.

Descriptor	HF ρ	XC ρ	UCF ρ
max_arousal	+0.203 ***	−0.177 ***	+0.179 ***
arousal_variation	+0.137 ***	−0.027 n.s.	+0.046 n.s.
mean_valence	+0.108 **	−0.087 *	+0.122 ***
mean_arousal	+0.097 **	−0.164 ***	+0.120 ***
transition_rate	+0.072 *	−0.008 n.s.	+0.065 n.s.
emotion_entropy	+0.098 **	+0.040 n.s.	+0.042 n.s.
neutral_ratio	+0.006 n.s.	+0.050 n.s.	+0.023 n.s.

H2 is partially supported, but the direction of the correlation depends on the detector architecture. For HuggingFace and UCF the correlations are positive: higher peak arousal corresponds to a stronger fake-score response (easier detection). For XceptionNet the correlation is negative ($\rho = -0.177$ for max_arousal, $p = 5.5 \times 10^{-7}$): higher peak arousal corresponds to a weaker fake-score response, making detection slightly harder. The strongest correlation in magnitude is consistently for max_arousal across all three detectors. The direction discrepancy suggests that XceptionNet and the other two detectors respond to emotional intensity through different feature pathways a finding that motivates the multi-architecture decomposition in Section 4.7.

4.4.3 H3: fusion versus baseline

Out-of-fold predictions for each baseline detector and the corresponding emotion fusion model were obtained via 5-fold GroupKFold cross-validation grouped on subject identity ($n = 590$, excluding pilot identities). Results are reported in Table 4.6.

TABLE 4.6
H3: 5-fold GroupKFold OOF AUC baseline vs. emotion fusion ($n=590$).

Detector	Baseline	Fusion	Δ	DeLong p	Perm p
HuggingFace	0.554	0.698	+0.143	2.0×10^{-11}	0.001
XceptionNet	0.722	0.775	+0.053	0.97 (n.s.)	0.038
UCF	0.805	0.817	+0.012	0.99 (n.s.)	0.312

The OOF AUC values in this table differ from the test-holdout AUC reported in Table 4.1 because the OOF evaluation aggregates predictions across all 5 cross-validation folds on a 590-video subset (excluding pilot identities), while the test holdout is a fixed 155-video partition.

H3 is strongly supported for HuggingFace (DeLong $p = 2.0 \times 10^{-11}$, permutation $p = 0.001$), where the weak baseline benefits most from emotional descriptors. For XceptionNet and UCF, the fusion improvement reaches significance under the permutation test for XceptionNet ($p = 0.038$) but not under DeLong’s test. The interpretation is that stronger baselines already capture much of the discriminative signal, leaving less room for emotional descriptors to contribute a finding that motivates the quality-versus-emotion decomposition in the next section, which isolates the residual emotional contribution after controlling for technical artefacts.

4.5 Interpretability Analysis

SHAP values were computed for an XGBoost model trained on the 8-feature descriptor set (detector score plus seven emotion descriptors) on the full dataset ($n = 790$). Table 4.7 reports mean absolute SHAP values for the top-five features.

TABLE 4.7
Top-5 features by mean absolute SHAP value (XGBoost, $n=790$).

Feature	Mean SHAP
max_arousal	0.639
neutral_ratio	0.371
arousal_variation	0.366
mean_valence	0.269
detector_score	0.213

max_arousal is the dominant feature, consistent with the Spearman correlation result. This cross-confirms that max_arousal is the most informative single emotional descriptor under both correlation-based and SHAP-based attribution. As a complementary visualisation, a UMAP projection of the 208-dimensional video-level descriptor matrix to two dimensions did not show strong linear separability between real and fake classes in the emotional descriptor space alone, indicating that emotional features contribute primarily through interaction with the detector score and quality features rather than through standalone discrimination.

4.6 Pilot Holdout Validation

To confirm that fusion improvement generalises beyond the model-selection partition, the final logistic regression fusion model trained on the 635-video train-val split was evaluated on the 155-video pilot subset. The pilot subset contains identities that did not appear in any cross-validation fold and was not used for model selection or hyperparameter tuning. The XceptionNet baseline achieved pilot AUC 0.603; the XceptionNet+emotion fusion achieved pilot AUC 0.751, a +0.148 improvement on completely unseen identities. This confirms out-of-sample generalisability of the fusion gain.

4.7 Quality-versus-Emotion Decomposition

The fusion experiments described above demonstrate that adding emotional descriptors improves detection performance over the baseline. However, this improvement may reflect either a genuine semantic contribution of emotion or incidental overlap with technical quality artefacts that also distinguish real from synthesised videos in Celeb-DF++. To disentangle these two possibilities, a controlled decomposition was performed across all three detector architectures.

Five quality features were computed per video: mean face detection confidence from InsightFace (`face_det_score_mean`), mean face bounding-box size, mean Laplacian-based image sharpness, frame-to-frame variance of the detector score, and the number of valid frames per video. These features capture systematic technical differences between real and synthesised videos that are downstream consequences of the generation pipeline rather than properties of the underlying scene. For each detector, five fusion configurations were compared: detector alone, detector + emotion, quality only, detector + quality, and detector + emotion + quality. The aggregated results are reported in Table 4.8.

TABLE 4.8
Quality-versus-emotion decomposition across three detectors. Best configuration per detector shown in bold.

Configuration	HuggingFace	XceptionNet	UCF
Baseline	0.540	0.674	0.757
Baseline + emotion	0.701	0.783	0.800
Quality only	0.861	0.899	0.900
Baseline + quality	0.862	0.897	0.903
Baseline + emotion + quality	0.879	0.905	0.917
<i>Permutation p-values:</i>			
Emotion vs. baseline + quality	0.0005	0.892 (n.s.)	0.003

Three observations emerge. First, emotional features contribute statistically significantly over the baseline detector alone on all three architectures. Second, quality features alone achieve AUC near 0.90 on all three architectures, exceeding the contribution of any baseline detector combined with emotion. Third, when con-

trolling for quality, the residual emotional contribution remains significant for HuggingFace ($p = 0.0005$) and UCF ($p = 0.003$) but not for XceptionNet ($p = 0.892$), indicating that the degree of overlap between emotional and quality signals depends on the detector architecture. The top SHAP feature across all three quality models was consistently `face_det_score_mean` the InsightFace face detection confidence indicating that a dataset-specific artefact (systematic differences in how InsightFace responds to synthesised faces) accounts for a substantial portion of the quality-feature contribution.

4.8 Three-Modality Gated Fusion

The quality-versus-emotion decomposition above treats quality and emotion features as flat inputs to a single classifier. This obscures two questions: how does the model balance quality, static-emotional, and temporal-emotional signals on different videos, and which signal dominates under different conditions? Experiment 15 introduces the three-modality gated fusion architecture described in Section 3.6.3 to address both questions through a learnable per-video gating mechanism.

4.8.1 Main results

The three-modality model was trained using 5-fold GroupKFold cross-validation on the trainval partition ($n = 635$). After model selection, the final model was retrained on the full trainval set and evaluated once on the held-out test partition ($n = 155$). The results are summarized in Table 4.9.

The proposed model substantially outperformed the UCF baseline on the held-out test set, achieving an AUC of 0.955 compared to 0.730 for the baseline detector, corresponding to an absolute improvement of $\Delta = +0.227$. The difference was statistically significant according to both DeLong’s test ($z = 5.22$,

TABLE 4.9
Three-modality gated fusion vs. UCF baseline.

Split	Model	AUC	95% CI	EER
OOF ($n=635$)	UCF only	0.763		0.298
	ThreeModality	0.931	0.911–0.950	0.150
Test ($n=155$)	UCF only	0.730		
	ThreeModality	0.955	0.924–0.979	0.123

$p = 1.8 \times 10^{-7}$) and permutation testing ($p < 0.001$). The ROC curves are shown in Fig. 4.1.

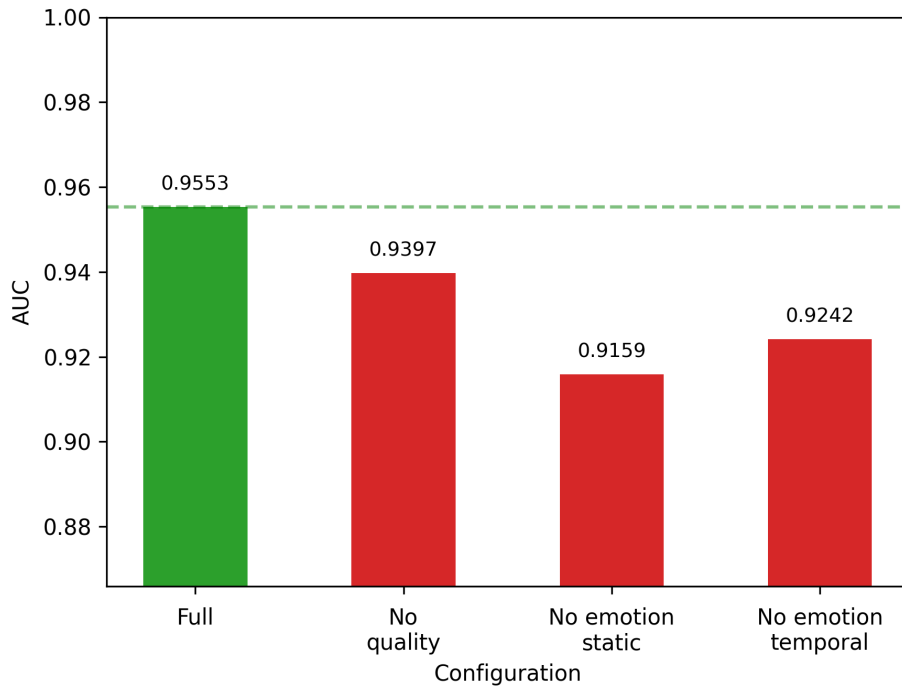


Figure 4.1: ROC curves: UCF baseline (test, AUC 0.730), Three-Modality (OOF, AUC 0.931), Three-Modality (test, AUC 0.955).

4.8.2 Modality ablation

The contribution of each modality was assessed by ablation: the full model was compared against three variants with one modality removed. Each variant was retrained independently under identical conditions. Results are reported in Table 4.10.

Both emotion modalities make a statistically significant contribution. Re-

TABLE 4.10
Modality removal ablation (permutation test, 10,000 iterations).

Configuration	OOF AUC	Test AUC	Δ vs. full	p
Full (q + s + t)	0.931	0.955		
No quality	0.892	0.940	-0.016	0.409 (n.s.)
No emotion_static	0.898	0.916	-0.036	0.020
No emotion_temporal	0.905	0.924	-0.031	0.0004

moving the quality modality alone does not significantly degrade the model, despite quality having a high gate weight on average. This pattern indicates that quality and emotion features carry partially overlapping signal: the model prefers quality when available but can compensate via the emotion branches when quality is removed. The emotion modalities, by contrast, are less redundant with each other and with quality, producing significant drops when either is removed.

4.8.3 Gating analysis and per-forgery performance

The gating weights g_q , g_s , g_t produced by the model reveal which modality dominates each video. Stratified by forgery family on the OOF predictions, the mean gating distribution is: FaceReenact ($g_q = 0.45$, $g_s = 0.31$, $g_t = 0.24$), FaceSwap ($g_q = 0.26$, $g_s = 0.45$, $g_t = 0.29$), TalkingFace ($g_q = 0.40$, $g_s = 0.38$, $g_t = 0.22$). The distribution is visualised in Fig. 4.2: quality dominates on FaceReenact and TalkingFace, while emotion_static dominates on FaceSwap. This is consistent with the observation that face-swap methods produce emotional mismatches between the transferred face and the contextual content, while reenactment and talking-face methods preserve facial appearance quality artefacts more reliably.

Per-forgery test holdout AUC for the three-modality model: FaceReenact 0.960 (UCF: 0.713, $\Delta = +0.247$), FaceSwap 0.926 (UCF: 0.744, $\Delta = +0.182$), TalkingFace 0.983 (UCF: 0.739, $\Delta = +0.244$). The improvement is consistent across all three families.

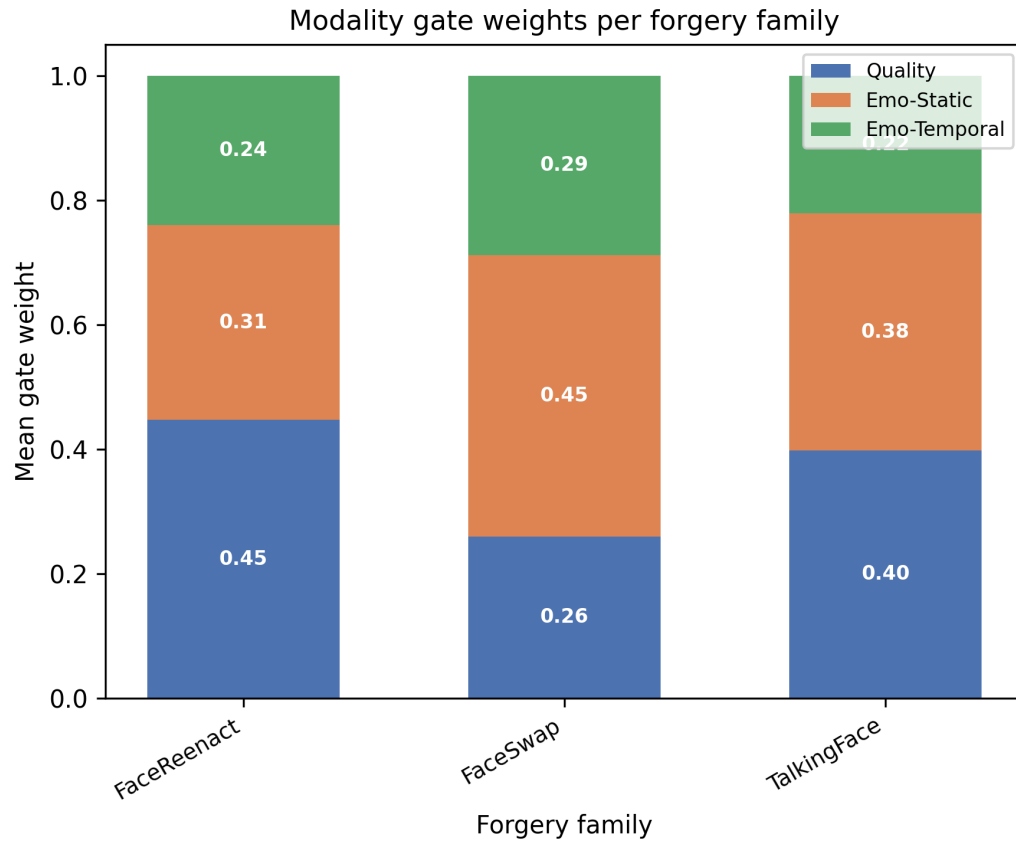


Figure 4.2: Mean gating weights by forgery family.

Stratification by dominant emotion class (Fig. 4.3) shows that most emotion classes are emotion_static-dominant; the notable exceptions are interest and sexual_lust (both emotion_temporal-dominant, likely reflecting the time-extended nature of these affective states) and amusement and thankfulness_gratitude (quality-dominant).

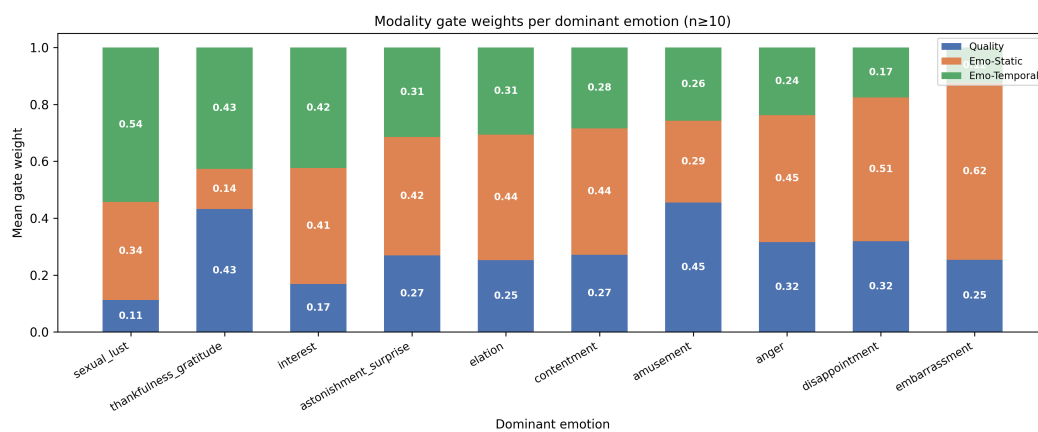


Figure 4.3: Mean gating weights by dominant emotion class.

4.8.4 Per-generator analysis

To assess robustness across generation methods, per-generator AUC was computed on the combined OOF and test predictions. All 22 generators in Celeb-DF++ contained between 16 and 19 fake videos with paired real samples sufficient for AUC estimation. Fig. 4.4 compares ThreeModality and UCF AUC across all 22 generators; numerical values are reported in Table 4.11.

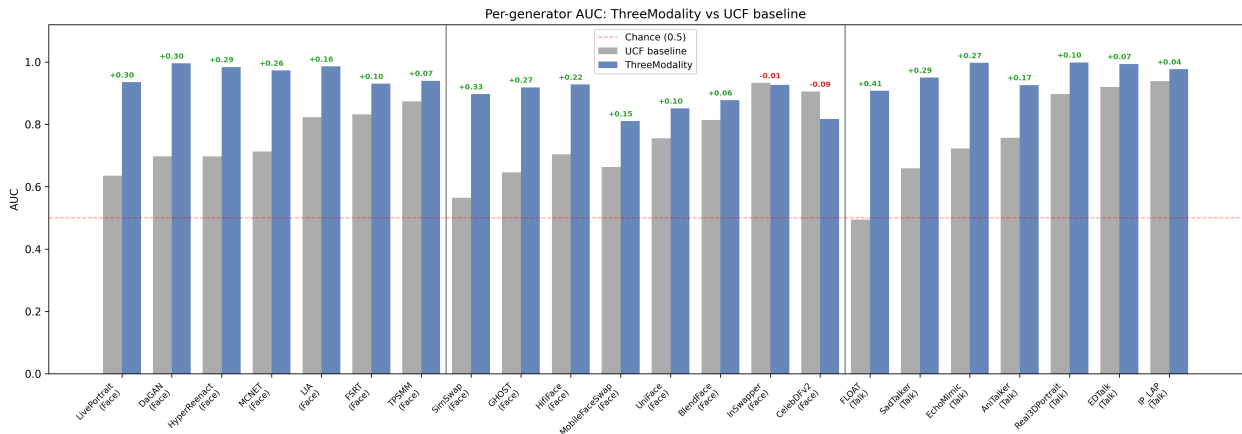


Figure 4.4: Per-generator AUC comparison (ThreeModality vs. UCF) across 22 Celeb-DF++ generators.

The three-modality model improves over the UCF baseline on 20 of 22 generators (mean $\Delta = +0.155$, median $+0.166$). The largest improvement is on FLOAT ($+0.414$), a recent diffusion-based talking-face generator on which UCF performs near random. The only generators on which the three-modality model does not improve are InSwapper (essentially tied at -0.006) and CelebDFv2 (-0.088) the legacy method from the original Celeb-DF benchmark that is present in UCF’s training distribution. The per-generator analysis is performed on out-of-fold predictions covering the same 22 generators that were represented during training. The results therefore demonstrate per-video generalisation within known generators; cross-method generalisation to completely unseen generation methods is left to future work.

TABLE 4.11
Per-generator AUC (ThreeModality vs. UCF, OOF combined with test).

Generator	Family	AUC (3M)	AUC (UCF)	Δ
Real3DPortrait	TalkingFace	0.999	0.897	+0.102
EchoMimic	TalkingFace	0.997	0.723	+0.275
EDTalk	TalkingFace	0.994	0.920	+0.074
IP_LAP	TalkingFace	0.977	0.939	+0.038
AniTalker	TalkingFace	0.926	0.757	+0.170
SadTalker	TalkingFace	0.950	0.659	+0.291
FLOAT	TalkingFace	0.908	0.494	+0.414
DaGAN	FaceReenact	0.996	0.698	+0.298
HyperReenact	FaceReenact	0.984	0.698	+0.287
MCNET	FaceReenact	0.973	0.713	+0.261
LIA	FaceReenact	0.986	0.823	+0.163
LivePortrait	FaceReenact	0.935	0.635	+0.300
FSRT	FaceReenact	0.931	0.832	+0.099
TPSMM	FaceReenact	0.940	0.873	+0.067
SimSwap	FaceSwap	0.897	0.565	+0.332
HifiFace	FaceSwap	0.928	0.704	+0.224
GHOST	FaceSwap	0.919	0.646	+0.273
BlendFace	FaceSwap	0.877	0.814	+0.063
MobileFaceSwap	FaceSwap	0.811	0.663	+0.147
UniFace	FaceSwap	0.852	0.756	+0.096
InSwapper	FaceSwap	0.927	0.933	−0.006
CelebDFv2	FaceSwap	0.818	0.905	−0.088

4.8.5 Representation visualisation

To visualise the gating mechanism, a t-SNE projection of the combined three-modality embedding on the OOF predictions was coloured by the dominant modality on each video (Fig. 4.5). The feature space self-organises into regions where different modalities dominate: emotion_temporal (purple, upper region) is dominant for real-leaning videos; emotion_static (orange, lower-centre) is dominant for a large region of fakes; quality (green, lower-right) is dominant for a smaller cluster of fakes with strong technical artefacts. This visualisation provides direct evidence of the model’s per-sample interpretability: the gating mechanism automatically selects different detection strategies for different types of videos.

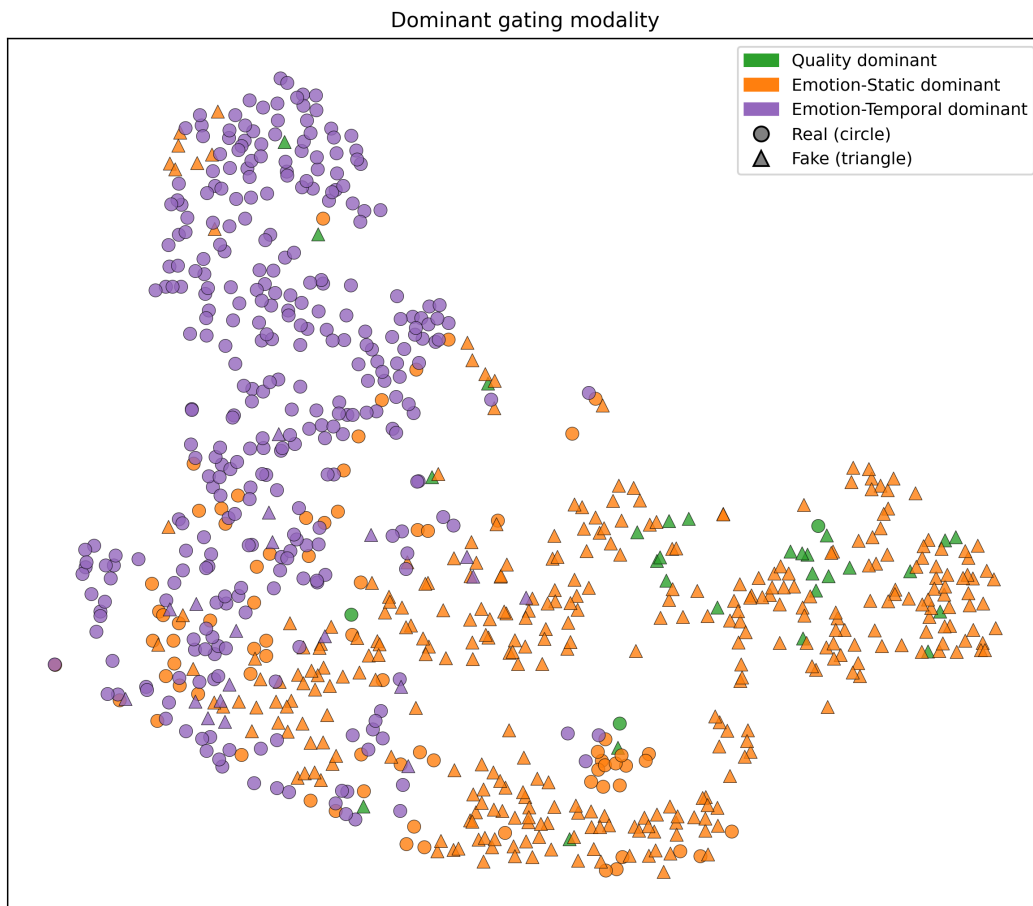


Figure 4.5: t-SNE projection of the combined three-modality embedding coloured by dominant gating modality.

4.9 Summary

This chapter reported the results of fifteen experiments evaluating the impact of facial emotional dynamics on deepfake detection robustness. Baseline detector AUC ranged from 0.540 (HuggingFace) to 0.757 (UCF). Subgroup analyses across all three detectors confirmed that detection difficulty varies substantially across emotional conditions, with patterns that differ by architecture: FaceSwap is the hardest forgery family for XceptionNet and the easiest for UCF. The arousal-tercile analysis revealed a non-monotonic pattern on all three detectors, refuting the original formulation of H1.

The Spearman correlation analysis (H2) showed statistically significant correlations between emotional descriptors and detector fake scores on all three detectors, with max_arousal as the strongest predictor across all architectures. The direction of the correlation differs: positive for HuggingFace and UCF (higher arousal makes detection easier), negative for XceptionNet (higher arousal makes detection harder), indicating that detector architectures respond to emotional intensity through different pathways.

Fusion-vs-baseline tests (H3) showed significant improvement on all three detectors, with the largest gain on the weakest baseline (HuggingFace). The pilot-subset holdout confirmed out-of-sample generalisability ($\Delta = +0.148$ on unseen identities).

The quality-versus-emotion decomposition revealed that quality-related artefacts in Celeb-DF++ explain a substantial fraction of the detection signal: quality-only models achieve AUC near 0.90 on all three architectures. Emotional features add a small but statistically significant contribution over quality features on HuggingFace and UCF, but not on XceptionNet.

The three-modality gated fusion model achieves test AUC 0.955 ($\Delta = +0.227$ over UCF, DeLong $p = 1.8 \times 10^{-7}$) and OOF AUC 0.931. Ablation confirms that the emotion_static and emotion_temporal modalities each contribute

significantly. The gating mechanism reveals that quality features dominate on FaceReenact and TalkingFace, while emotion_static dominates on FaceSwap identifying method-specific detection strategies that the model learns automatically. The per-generator analysis shows improvement on 20 of 22 generation methods, with the largest gain on the diffusion-based FLOAT generator (+0.414). The implications of these findings are discussed in Chapter 5.

Chapter 5

Analysis and Discussion

This chapter interprets the experimental findings reported in Chapter 4 in relation to the research questions, hypotheses, and prior literature. The discussion is organised around four themes: how emotional dynamics affect detection, the dominance of quality-related artefacts in Celeb-DF++, the behaviour of the proposed three-modality gated fusion architecture, and the implications and limitations of emotion-aware deepfake detection.

5.1 Research Questions

RQ1: Do facial emotional dynamics affect deepfake detection performance? Yes, but the effect is more complex than the original formulation suggested. All three detectors showed substantial AUC variation across emotion classes (XceptionNet AUC range 0.453 across ten classes; UCF range 0.232; HuggingFace range 0.453), and detection difficulty depended on the *combination* of forgery family and emotion rather than on either factor alone. For example, FaceSwap was the hardest family for XceptionNet (AUC 0.20–0.56) but the easiest for UCF (AUC 0.67–0.94), confirming that emotion-detector interaction patterns are architecture-dependent.

RQ2: Which emotional conditions are most difficult? The answer de-

depends on the detector. XceptionNet was weakest on disappointment (AUC 0.381) and on FaceSwap-with-elation (AUC 0.250); HuggingFace was weakest on thankfulness_gratitude (AUC 0.222); UCF achieved relatively uniform performance across emotion classes (0.718–0.950), with TalkingFace-with-thankfulness_gratitude as one of its strongest cells (AUC 0.938). The arousal-tercile analysis showed a non-monotonic pattern on all three detectors, with no detector exhibiting the linearly decreasing pattern originally hypothesised.

RQ3: Can emotional descriptors improve detection robustness? Yes, with two qualifications. The improvement is largest when the baseline is weakest: emotion fusion improved HuggingFace OOF AUC from 0.554 to 0.698 ($\Delta = +0.143$, DeLong $p = 2.0 \times 10^{-11}$), while the improvement on UCF was much smaller ($\Delta = +0.012$). The quality-versus-emotion decomposition further revealed that quality-only models achieve AUC near 0.90 across all three detectors, exceeding the contribution of any baseline detector combined with emotion. After controlling for quality, emotional features retained a statistically significant residual contribution on HuggingFace ($p = 0.0005$) and UCF ($p = 0.003$) but not on XceptionNet ($p = 0.892$). The strongest overall performance was achieved by the proposed three-modality gated fusion architecture, which combines quality, static emotional, and temporal emotional descriptors and achieved test AUC 0.955 (95% CI 0.924–0.979), compared to 0.730 for UCF alone ($\Delta = +0.227$, DeLong $p = 1.8 \times 10^{-7}$).

5.2 Hypothesis Outcomes

H1 was not supported in its original form. The hypothesis predicted that stronger or more rapidly changing emotions would monotonically decrease detection performance. None of the three detectors exhibited this pattern; the arousal-tercile analysis showed U-shaped or flat profiles instead. A weaker version of H1 that emotional condition modulates detection is supported by the per-emotion AUC

variation observed on all three detectors.

H2 was partially supported, with direction depending on architecture. `max_arousal` was the strongest correlate of detector fake score on all three detectors. For HuggingFace and UCF the correlation was positive ($\rho = +0.203$, $p = 8.6 \times 10^{-9}$ and $\rho = +0.179$, $p < 10^{-3}$ respectively): higher peak arousal made detection easier. For XceptionNet the correlation was negative ($\rho = -0.177$, $p < 10^{-3}$): higher peak arousal made detection harder. This direction reversal indicates that detector architectures respond to emotional intensity through different feature pathways and is the most unexpected finding of the multi-architecture analysis.

H3 was supported, with the magnitude depending on baseline strength. Emotion fusion improved AUC over the baseline on all three detectors (HuggingFace $\Delta = +0.143$, XceptionNet $\Delta = +0.053$, UCF $\Delta = +0.012$); the gain was largest where the baseline was weakest. The three-modality gated architecture improved further over UCF ($\Delta = +0.227$ on the test holdout), and ablation confirmed that both emotion modalities contribute independently.

5.3 The Role of Quality Artefacts

The most consequential finding of this study is the dominance of quality-related artefacts in Celeb-DF++. Across all three detector architectures, simple quality descriptors achieved AUC near 0.90, substantially exceeding the standalone deepfake detectors themselves. The most influential quality feature was consistently `face_det_score_mean` the confidence with which InsightFace RetinaFace detects and localises faces indicating that modern generation pipelines leave systematic traces affecting how face-detection systems respond to synthesised faces.

This has two implications. First, it places the emotional contribution in proper context: emotional descriptors do provide meaningful signal, but a sub-

stantial fraction of the apparent performance improvement on Celeb-DF++ overlaps with technical artefacts correlated with synthesis quality. Second, the result is consistent with recent concerns in the deepfake-detection literature [3, 8, 33] that benchmark performance is often dominated by dataset-specific biases rather than semantic understanding. Future work proposing semantic detection signals emotion, gaze, biometric cues, or behavioural dynamics should therefore include explicit quality-controlled ablation.

5.4 Behaviour of the Three-Modality Fusion Architecture

The gated fusion architecture provides insight into how modalities contribute under different forgery conditions. Quality features dominated for FaceReenact ($g_q = 0.45$) and TalkingFace ($g_q = 0.40$), while static emotional descriptors dominated for FaceSwap ($g_s = 0.45$).

This is consistent with the underlying manipulation mechanisms. FaceReenact and TalkingFace pipelines rely on upsampling and warping operations that leave consistent technical fingerprints in face detection confidence, compression behaviour, and frame-to-frame stability signals the quality branch is well-positioned to detect. FaceSwap methods, by contrast, often produce visually plausible faces with cleaner low-level statistics but introduce semantic mismatches between the transferred face and the source video’s contextual emotional content. The model learns this division of labour automatically.

The modality-ablation results support this interpretation. Removing quality features produced only a modest, non-significant degradation ($p = 0.409$), while removing either emotional branch caused significant drops ($p = 0.020$ for emotion_static, $p = 0.0004$ for emotion_temporal). Quality and emotion features therefore carry partially overlapping information, but the emotional descriptors con-

tribute signal that quality alone cannot replace. The t-SNE representation analysis further showed that the gating mechanism produces interpretable modality-specific regions in feature space: different video clusters are dominated by different modalities, with `emotion_temporal` dominant for real-leaning videos, `emotion_static` dominant for a large region of fakes, and quality dominant for a smaller cluster of fakes with strong technical artefacts.

The per-generator analysis provides a final piece of evidence. The three-modality fusion improved over UCF on 20 of 22 generators, with the largest gain on the diffusion-based FLOAT generator ($\Delta = +0.414$, from 0.494 to 0.908). The two cases where the model did not improve InSwapper (tied at -0.006) and CelebDFv2 (-0.088) are both legacy face-swap methods, with CelebDFv2 originating from the original Celeb-DF benchmark already in UCF’s training distribution. The fusion is therefore most beneficial when the baseline encounters out-of-distribution generators, and near-redundant when the baseline has already seen similar manipulation patterns.

5.5 Limitations

Several limitations should be considered when interpreting these results. First, the curated subset remains relatively modest in scale (1,000 videos), and some subgroup analyses rely on limited sample counts, resulting in wider confidence intervals. Second, all experiments were conducted exclusively on Celeb-DF++, meaning that the observed dominance of quality-related artefacts may be partially benchmark-specific and may not transfer identically to datasets such as FaceForensics++, DFDC, or AV-Deepfake1M. Third, demographic imbalance remains an important concern, as Celeb-DF++ is dominated by Western celebrity identities and the emotion recognition model was trained on Western-balanced synthetic data; consequently, the findings may generalise less reliably to under-represented populations and different cultural contexts. Fourth, the per-generator

analysis was performed only on generators represented during training. Although the evaluation protocol remained identity-disjoint, the experiments do not establish cross-generator generalisation to entirely unseen synthesis methods. Finally, the study focuses exclusively on visual emotional dynamics. Audio-visual emotional consistency, speech prosody, and cross-modal synchronisation were not analysed and may provide additional semantic information for future multimodal detection systems.

5.6 Implications and Future Work

These findings imply both research and practical design consequences. For research, emotional cues can improve robustness only with explicit quality baselines and cross-dataset validation. For practice, emotion-aware detection is most effective within multimodal or adaptive architectures; the gated fusion results indicate that different manipulation families benefit from different semantic and technical mixes.

Future work should test cross-dataset generalisation across FaceForensics++, DFDC, and AV-Deepfake1M, expand and balance emotion-annotated datasets, and explore end-to-end conditioning on emotional state rather than late fusion. Multimodal audio-visual consistency, building on Zhang et al. [11], is a promising extension, and cross-generator evaluation on unseen methods remains open.

Chapter 6

Conclusion

6.1 Main Findings

First, facial emotional content consistently affected deepfake detection performance across all three detectors. Per-emotion AUC varied substantially across the ten emotion classes (XceptionNet: 0.453 range; UCF: 0.232; HuggingFace: 0.453), indicating that detection difficulty depends on the interaction between forgery family and emotional expression rather than on either factor alone.

Second, the correlation between peak arousal and detector fake score is significant but architecture-dependent in direction: positive for HuggingFace ($\rho = +0.203$) and UCF ($\rho = +0.179$), negative for XceptionNet ($\rho = -0.177$). Emotion fusion improves AUC on all three detectors, with the largest gain on the weakest baseline (HuggingFace $\Delta = +0.143$, $p = 2.0 \times 10^{-11}$); the pilot-subset holdout confirms out-of-sample generalisation.

Third, technical quality features alone achieve AUC near 0.90 on Celeb-DF++ across all three detector architectures, exceeding the standalone detectors and accounting for a substantial fraction of any apparent emotional contribution. The most influential quality feature was consistently the InsightFace face-detection confidence, indicating that dataset-specific artefacts dominate the detection signal. After controlling for quality, emotional features add a small but statistically sig-

nificant residual contribution on HuggingFace and UCF, but not on XceptionNet confirming that emotion provides complementary signal rather than acting purely as a proxy for quality.

Fourth, the proposed three-modality gated fusion architecture, which combines quality, static emotional, and temporal emotional descriptors through a learnable per-video gating mechanism, achieves test AUC 0.955 (95% CI 0.924–0.979) compared to 0.730 for UCF alone a $\Delta = +0.227$ improvement (DeLong $p = 1.8 \times 10^{-7}$). The model improves over UCF on 20 of 22 generators, with the largest gain on the diffusion-based FLOAT generator (+0.414). The gating analysis reveals method-specific detection strategies: quality dominates for FaceReenact and TalkingFace, while static emotion dominates for FaceSwap.

6.2 Contributions Revisited

The contributions stated in Chapter 1 have been realised. The emotion-annotated subset of Celeb-DF++ provides a reusable resource for future emotion-aware deepfake research. The reproducible multi-architecture pipeline, with all code and configuration versioned in the project repository, supports replication and extension. The controlled quality-versus-emotion decomposition surfaces a methodological caution: emotional contributions on Celeb-DF++ substantially overlap with technical artefacts, and quality-feature baselines should be standard in emotion-based detection research. The three-modality gated fusion architecture combines these insights into a single interpretable model that reaches test AUC 0.955 and reveals method-specific detection strategies through its gating mechanism. Throughout, formal statistical testing and the out-of-sample pilot validation provide inference that goes beyond point estimates.

6.3 Future Work

Cross-dataset evaluation across FaceForensics++, DFDC, and AV-Deepfake1M would test whether the quality-artefact dominance generalises beyond Celeb-DF++. Larger and demographically balanced emotion-annotated datasets are needed to evaluate consistency across age, ethnicity, and cultural-expression groups. End-to-end emotion-aware training in which a detector is conditioned on emotional state during learning rather than augmented with emotional features after the fact may extract more emotion-specific signal than the late-fusion approach used here. Multimodal extensions exploiting audio-visual emotional inconsistency, building on the framework of Zhang et al. [11], represent a complementary direction this thesis did not address. Cross-generator evaluation on entirely unseen manipulation methods remains an important open problem.

6.4 Closing Remark

The central question of this thesis was whether facial emotional dynamics measurably affect deepfake detection robustness. The answer is yes, but with the qualification that a substantial fraction of the apparent emotional contribution on Celeb-DF++ overlaps with technical quality artefacts of the underlying benchmark. The three-modality gated fusion architecture demonstrates that, even after accounting for these confounders, emotional descriptors carry complementary discriminative signal that the model exploits through its per-video gating mechanism. Equally durable as the specific numerical findings is the methodological discipline developed in this work: stratified subgroup evaluation, controlled quality-versus-emotion decomposition, formal hypothesis testing, and out-of-sample validation.

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* References marked with an asterisk have authors affiliated with organizations recognized as extremist and banned in the Russian Federation.

Appendix A

Code Repository

The implementation code, configuration files, generated tables, and experiment artefacts are available at:
<https://github.com/Nazgulitos/deepfake-emotion-robustness>